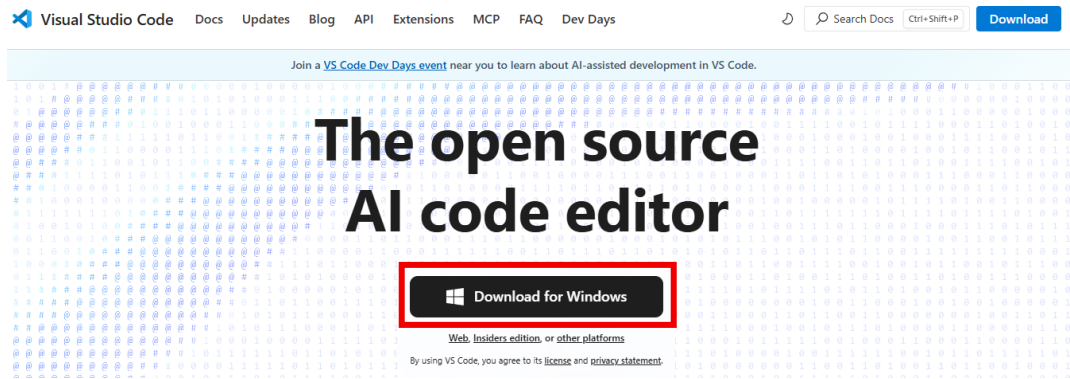


LG Prompthon Challenge 2025 Hands on Lab

실습 환경 생성

1. VSCode 설치

- [Visual Studio Code - Code Editing. Redefined](#) 접속
- Download for Windows 클릭하여 다운로드

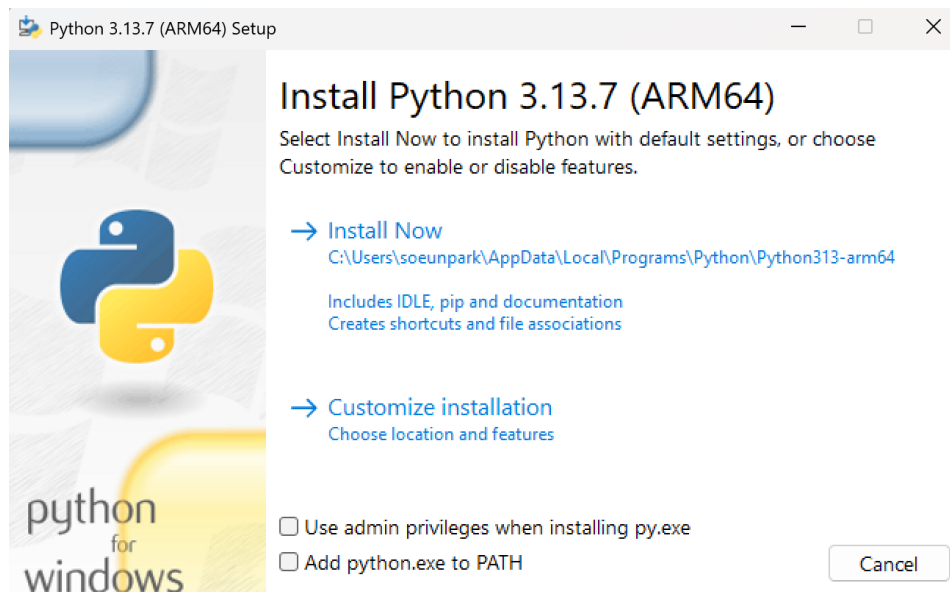


2. Python 설치

- [Welcome to Python.org](#)
- Python 버전 선택 후 (3.13.7 권장) OS 에 맞는 installer 다운로드하여 설치

Files							
Version	Operating System	Description	MD5 Sum	File Size	GPG	Sigstore	SBOM
Gzipped source tarball	Source release		138c2e19c835ead10499571e0d4cf189	28.0 MB	SIG	.sigstore	SPDX
XZ compressed source tarball	Source release		256cdb3bbf45cdce7499e52ba6c36ea3	21.7 MB	SIG	.sigstore	SPDX
macOS 64-bit universal2 installer	macOS	for macOS 10.13 and later	ac0421b04eef155f4daab0b023cf3956	67.8 MB	SIG	.sigstore	
Windows installer (64-bit)	Windows	Recommended	1da92e43c79f3d1539dd23a3c14bf3f0	27.5 MB	SIG	.sigstore	SPDX
Windows installer (32-bit)	Windows		f501c1b321c82412ed330ec5604cac39	26.2 MB	SIG	.sigstore	SPDX
Windows installer (ARM64)	Windows	Experimental	66c0ba98b20b7d4ea0904223d484d369	26.8 MB	SIG	.sigstore	SPDX
Windows embeddable package (64-bit)	Windows		77f294ec267596827a2ab06e8fa3f18c	10.4 MB	SIG	.sigstore	SPDX
Windows embeddable package (32-bit)	Windows		53eae0de1231fd133ea703e3e43b30	9.2 MB	SIG	.sigstore	SPDX
Windows embeddable package (ARM64)	Windows		6a3fcfc7a10a3822459b1c214f769df1	9.7 MB	SIG	.sigstore	SPDX
Windows release manifest	Windows	Install with 'py install 3.13'	6a5588452e73f959c9fa1ba009f42775	14.6 KB		.sigstore	

- 설치 시 Add python.exe to PATH 선택

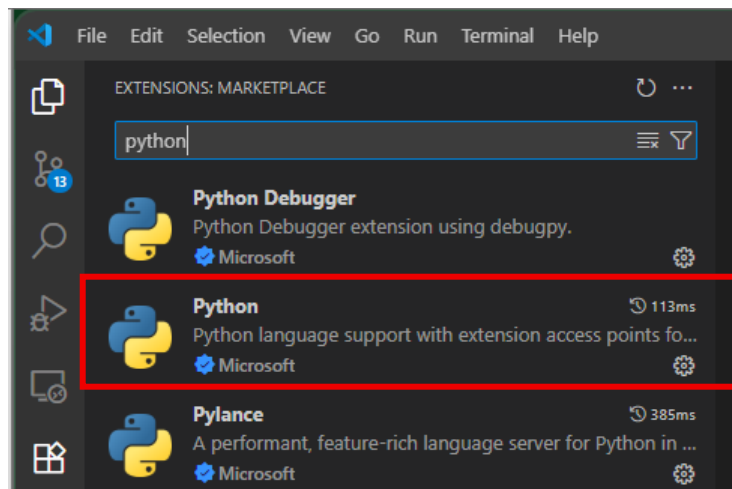


3. VSCode extension 설치

- 왼쪽 메뉴에서 아래 아이콘 클릭



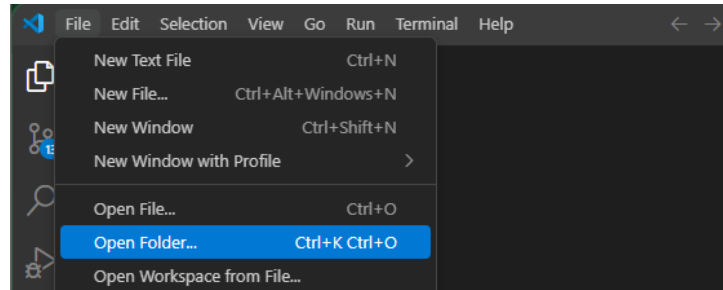
- Python 선택 후 install



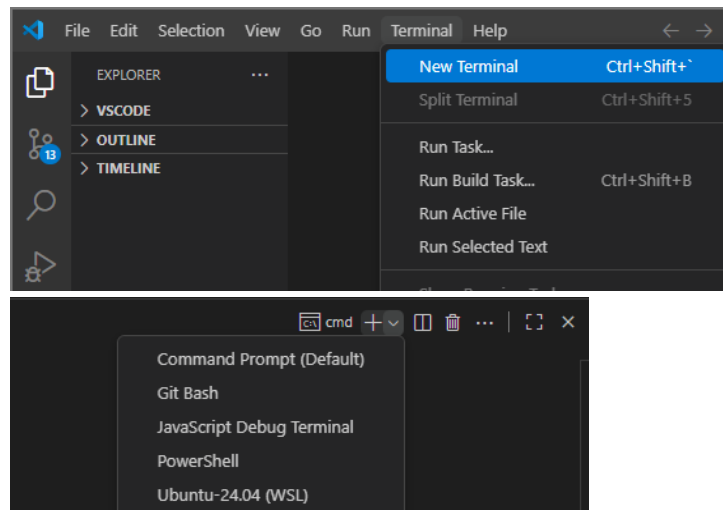
4. Python venv 설치

- 프로젝트 폴더 설정 (File > Open Folder)

※ 기존 폴더를 지정해도 되지만, 새로운 폴더 생성 후 사용을 권장합니다.



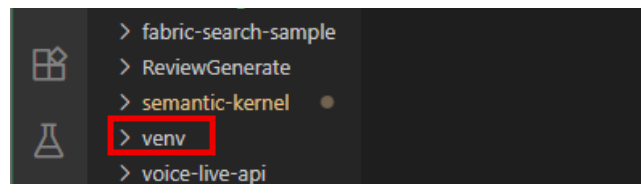
- New Terminal> cmd (Command Prompt) 선택



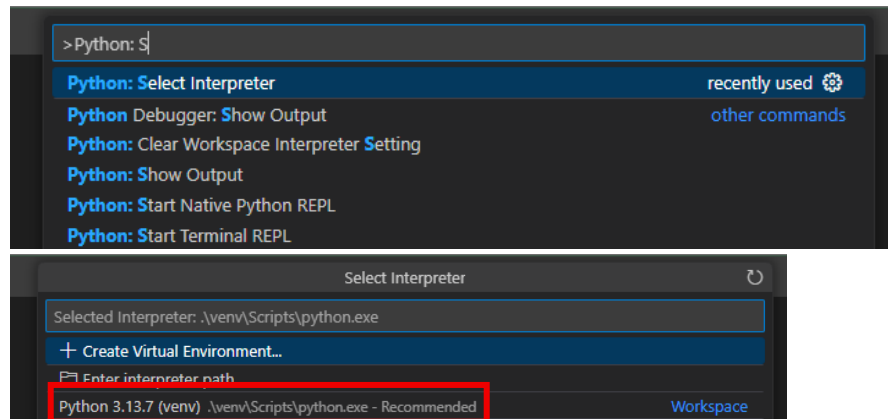
- Terminal 에서 아래 명령어 실행

- `python -m venv venv`

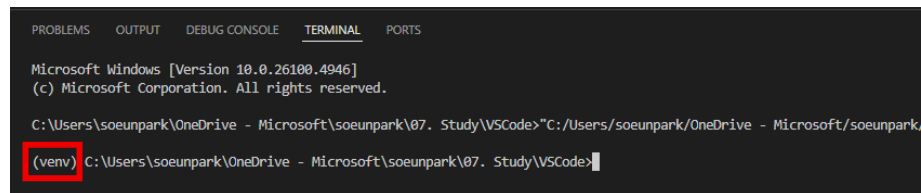
- venv 생성 확인



- [Ctrl + Shift + P] 키를 눌러 Python: Select Interpreter 검색하여 생성한 venv 선택



- Terminal 재실행 > venv 활성화 확인

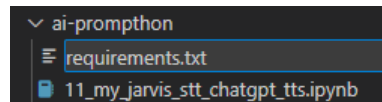


5. 실습에 필요한 패키지 설치

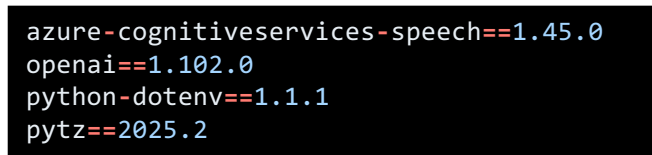
- Explorer 탭에서 New File 아이콘 선택



- requirements.txt 생성



- requirements.txt 에 아래 내용 붙여 넣고 저장



- requirements.txt 설치
 - Terminal 에서 아래 명령어 실행
 - `pip install -r requirements.txt`

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Users\soeunpark\OneDrive - Microsoft\soeunpark\07. Study\VSCode>"C:/Users/soeunpark/OneDrive - Microsoft
(venv) C:\Users\soeunpark\OneDrive - Microsoft\soeunpark\07. Study\VSCode>pip install -r requirements.txt

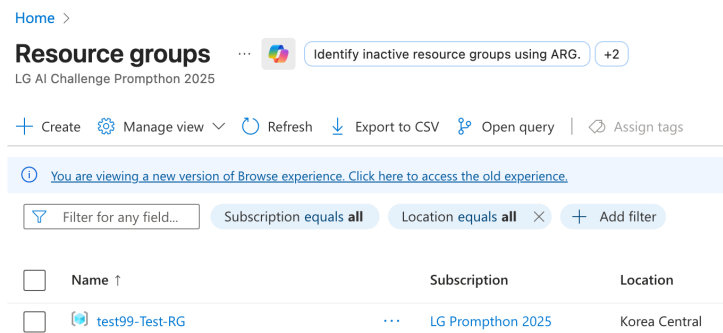
```

기본 환경 설정 및 권한 확인

1. Azure portal 접속
 - [Home - Microsoft Azure](https://portal.azure.com) (portal.azure.com)
2. Resource group 생성 확인
 - 아래 아이콘 클릭하여 리소스 그룹 접근



3. 권한 확인
 - 생성된 Resource groups 클릭

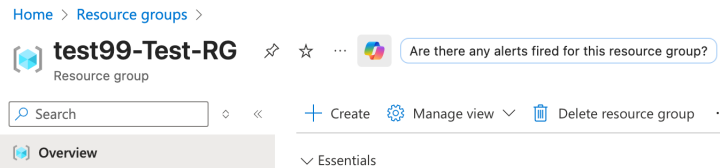


#1 AI Foundry 실습

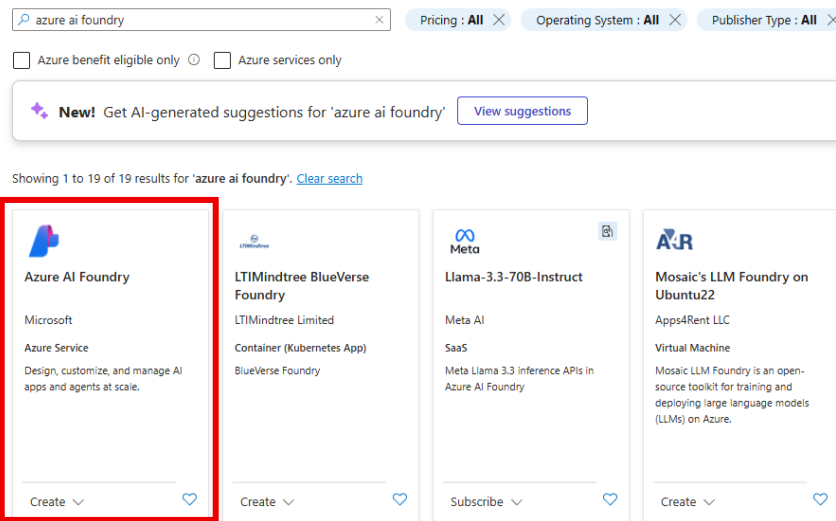
AI Foundry 프로젝트를 생성하고 간단한 RAG 기반 agent 생성

1. AI Foundry 리소스 생성

- 리소스 그룹 선택 > Create 클릭



- 검색창에 Azure AI Foundry 검색 후 Create 클릭



- 설정 값 입력

[Basics]

Resource group: 생성된 리소스 그룹 선택

Name: <생성할 리소스 이름 입력>

Region: East US

※ Korea Central 선택도 가능하지만 일부 기능이 preview 로 지원되지 않아 East US 를 권장합니다.

- 나머지 설정은 그대로 두고 Next 클릭

- Review + submit 설정 내용 확인 > Create
Create an Azure AI Foundry resource ...

✓ Basics
✓ Network
✓ Identity
✓ Encryption
✓ Tags
6 Review + submit

[View automation template](#)

Basics

Subscription LG Prompthon 2025
 Resource group test99-Test-RG
 Name soeunpark-ai-foundry
 Region East US
 Default project name firstProject

Network

Inbound Access All networks, including the internet, can access this resource.

Identity

Identity type System assigned

- 생성된 리소스 확인 (Azure AI Foundry, Azure AI Foundry project)

☐	Name ↑	Type	Location
☐	 soeunpark-ai-foundry ...	Azure AI Foundry	East US
☐	 firstProject (soeunpark-ai-foundry/firstProject) ...	Azure AI Foundry project	East US

2. Grounding with Bing Search 리소스 생성

- 리소스 그룹 선택 > Create 클릭

[Home](#) > [Resource groups](#) >

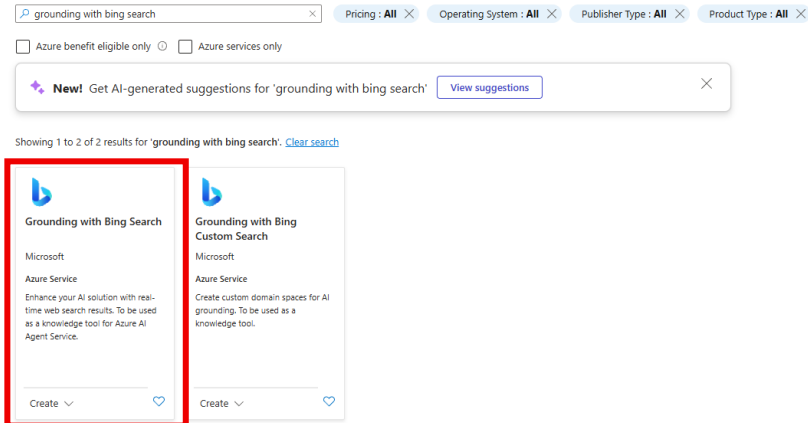

test99-Test-RG
☆ ☆ ...
 Are there any alerts fired for this resource group?

Resource group

◇ <<
+ Create
 Manage view ▾
 Delete resource group

 Overview
▽ Essentials

- 검색창에 Grounding with Bing Search 검색 후 Create 클릭



[Basics]

Resource group: 생성된 리소스 그룹 선택

Name: <생성할 리소스 이름 입력>

Region: Global

Pricing tier: Grounding with Bing Search (\$35K per 1K transactions)

- 나머지 설정은 그대로 두고 Next 클릭
- Review + submit 설정 내용 확인 > Create

Create a Grounding with Bing Search resource ...

Basics Tags **Review + create**

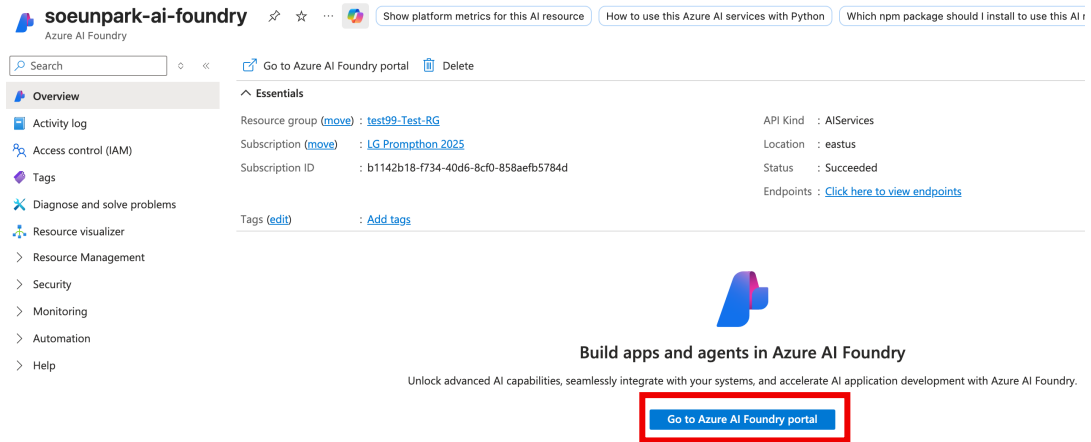
[View automation template](#)

Basics

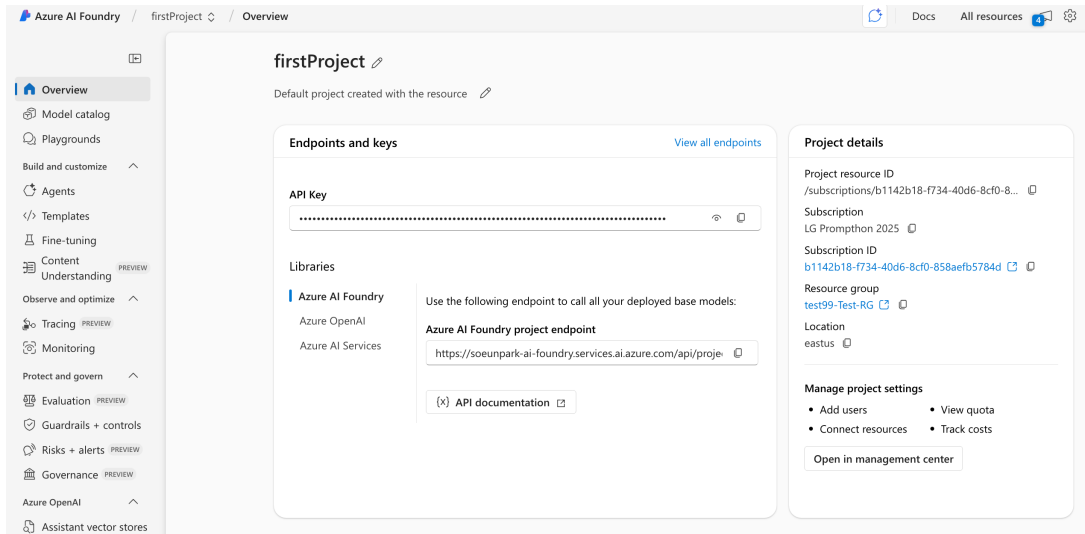
Subscription	LG Prompthon 2025
Resource group	test99-Test-RG
Name	soeunpark-bing-search
Resource group location	Korea Central
Pricing tier	Grounding with Bing Search (\$35 per 1K transactions)

3. AI Foundry portal 접속 (ai.azure.com)

- 생성된 Azure AI Foundry project 리소스 클릭 > Go to Azure AI Foundry portal

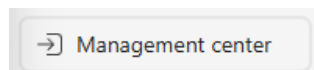


- AI Foundry portal 화면 확인

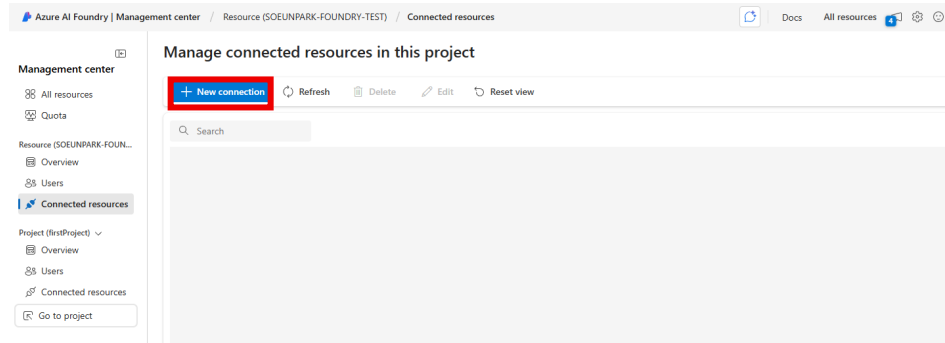


4. Grounding with Bing Search 연결

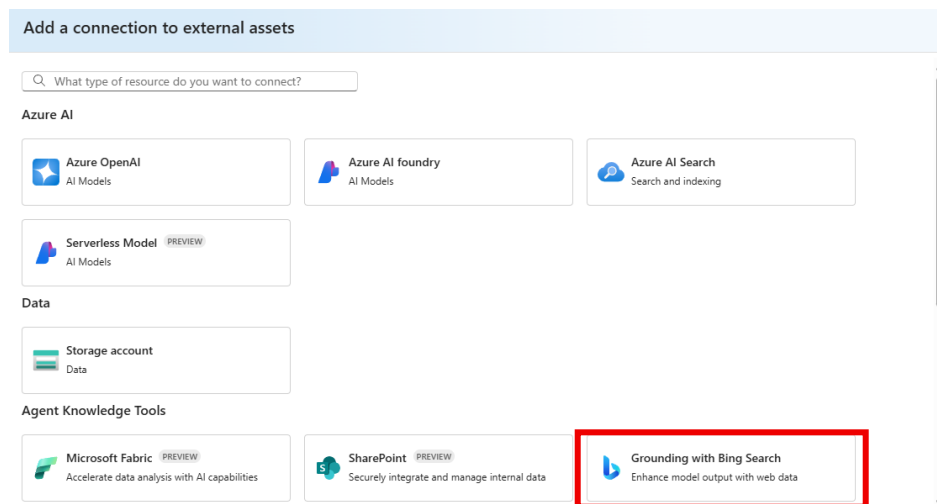
- 메뉴 설명 - All resources / Quota / Overview / Users / Connected resources
- Grounding with Bing Search 연결
 - Management center 클릭



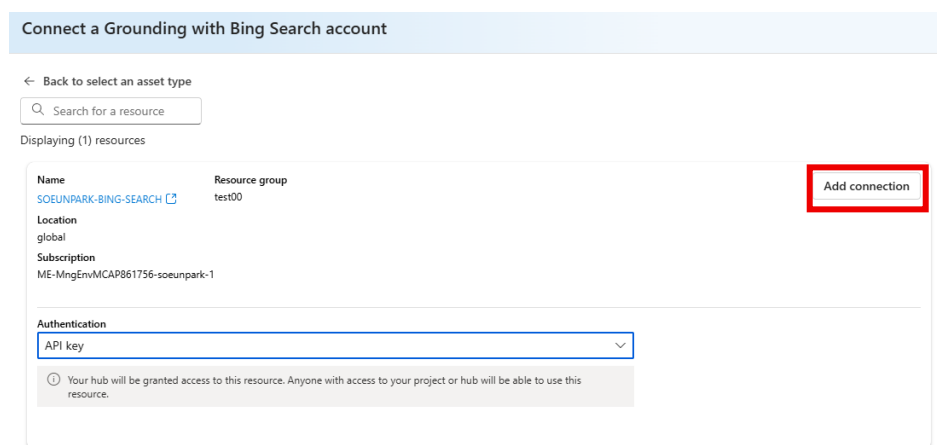
- Resource > Connected resources > New Connection



- Grounding with Bing Search 선택



- 기존에 생성했던 Grounding with Bing Search 확인 > Add connection



■ 연결 확인

Management center

- All resources
- Quota
- Resource (SOEUNPARK-FOUN...
- Overview
- Users
- Connected resources

Project (firstProject)

- Overview
- Users
- Connected resources

Go to project

Manage connected resources in this project

+ New connection
Refresh
Delete
Edit
Reset view

Search

Name	Access	Authentication type	Added by
SOEUNPARKBINGSEARCH	Shared to all projects	API key	soeunpark@microsoft.c...

■ 프로젝트로 돌아가기

Management center

- All resources
- Quota
- Resource (SOEUNPARK-FOUN...
- Overview
- Users
- Connected resources

Project (firstProject)

- Overview
- Users
- Connected resources

Go to project

Manage connected resources in this project

+ New connection
Refresh
Delete
Edit
Reset view

Search

Name	Access	Authentication type	Added by
SOEUNPARKBINGSEARCH	Shared to all projects	API key	soeunpark@microsoft.c...

5. AI Foundry 에 AOAI 모델 배포 – GPT-4.1, GPT-5

○ Model Catalog 탐색 – 모델 리스트, Benchmark

Azure AI Foundry / firstProject / Model catalog

Find the right model to build your custom AI solution
Models shown are available for deployment in this project. See all models

Announcements

Introducing Mistral Document AI (25.05)
Document processing with state-of-the-art OCR and structured data extraction
Check out model
Read blog

Introducing GPT-5 models
GPT-5 unifies frontier reasoning and advanced coding with high-performance...
Check out models
Read blog

Announcing new gpt-oss models
Push the open model frontier with gpt-oss-120k and gpt-oss-20k, released und...
Check out models
Read blog

Model Router with GPT-5
Model router fast-tracks efficient adoption of the GPT-5 models for your business.
Check out model
Read blog

FLUX.1 Kontext [pro]
Black Forest Labs foundat generation and editing m
Check out models

Model leaderboards
See what models are performing best in different criteria. Learn more

Quality
1 o3-pro
2 gpt-5
3 o3

Safety
1 Phi-4
2 o3
3 o1

Cost
1 Ministral-3B
2 Phi-4-mini-instruct
3 Phi-4-mini-reasoning

Throughput
1 grok-3-mini
2 llama-3.2-18-Instruct
3 o3-mini

Collections
Capabilities
Inference tasks
Fine-tuning tasks
Licenses

Search

Models 96

gpt-5-mini
Chat completion

gpt-5-nano
Chat completion

gpt-5-chat
Chat completion

FLUX.1.1-pro
Text to image

FLUX.1-Kontext-pro
Text to image, Image to image

codex-mini
Responses

DeepSeek-R1-0528
Chat completion

sora
Video generation

grok-3
Chat completion

grok-3-mini
Chat completion

model-router
Chat completion

o4-mini
Chat completion

MAI-D5-R1
Chat completion

gpt-4.1
Chat completion

gpt-4.1-mini
Chat completion

gpt-4.1-nano
Chat completion

mistral-medium-2505
Chat completion, Image classic...

Phi-4-reasoning
Chat completion

Phi-4-mini-reasoning
Chat completion

Llama-4-Scout-17B-16E-L
Chat completion

- gpt-4.1 클릭 > Use this model

gpt-4.1

▶ Use this model Fine-tune

Details Benchmarks Existing deployments License

The gpt-4.1 series is the latest iteration of the `gpt-4o` model family. This iteration of models is specifically targeted for better coding and instruction following, making it better at handling complex technical and coding problems.

In addition, it increases the context token limit up to 1M input tokens and provides separate billing for small context (128k) and large (up to 1M) context inputs.

- Customize > Tokens per Minute Rate Limit 500K 로 설정

Deployment details

Collapse

Model version upgrade policy

Upgrade once new default version becomes available

Model version

2025-04-14 (Default)

Connected AI resource

SOEUNPARK-FOUNDRY-EU2 (This resource)

730K tokens per minute quota available for your deployment

Tokens per Minute Rate Limit

500K

Corresponding requests per minute (RPM) = 500

Content filter

DefaultV2

Deploy

Cancel

- Deployment details 확인 > Deploy

Deploy gpt-4.1

Deployment name *

gpt-4.1

Deployment type

Global Standard

Global Standard: Pay per API call with the highest rate limits. Learn more about [Global deployment types](#).

Data might be processed globally, outside of the resource's Azure geography, but data storage remains in the AI resource's Azure geography. Learn more about [data residency](#).

Deployment details

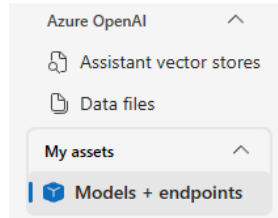
Customize

Model version	Authentication type
2025-04-14	Key
Capacity	Resource location
140K tokens per minute (TPM)	East US 2
Content safety	Version upgrade policy
DefaultV2	Once a new default version is available

Deploy

Cancel

- gpt-5 도 동일하게 배포
- 배포된 모델 확인
 - 왼쪽 메뉴에서 My Assets 밑에 Models + endpoints 선택



- 배포된 모델 리스트 확인

Model deployments Help

Model deployments Service endpoints

[+ Deploy model](#) [Refresh](#) [Edit](#) [Open in playground](#) [Reset view](#)

Name	Model name	Model version	State	Model retirement date	Content filter	Deployment type
This resource						
gpt-4.1	gpt-4.1	2025-04-14	Succeeded	Apr 11, 2026 9:00 AM	DefaultV2	Global Standard

6. Agent 생성

- Agents 메뉴 클릭 > agent 자동 생성

 A screenshot of the Azure OpenAI portal's 'Agents' page. A green success banner at the top reads 'Success: Agent with id asst_bE1ANTv9KNWlv4lInNxe0Gzst created successfully'. Below it, the 'Create and debug your agents' section shows 'My agents' and 'My threads' tabs. A '+ New agent' button and a 'Refresh' button are visible. A table lists the created agent:

Name	ID	Model	Created ↓	Description	Tools
Agent411	asst_bE1ANTv9KNWlv4lInNxe0Gzst	gpt-4.1	Aug 31, 2025 ...		

- Agent 설정

Agent name: <Agent 이름 입력>

Deployment: 이전에 배포한 gpt-4.1 모델 확인

Instructions: You are a helpful assistant. Search for information using the input that I give.

※ Instruction 예/output 형식 지정과 같이 agent 의 행동 설정 가능

예) Hi, Agent! Draw a graph for a line with a slope of 4 and y-intercept of 9.

Setup Hide

Try in playground

Agent ID ⓘ
asst_bE1ANTv9KNWlv4INxke0Gzst

Agent name
my-agent

Deployment * + Create new deployment
gpt-4.1 (version:2025-04-14)

Instructions ⓘ
You are a helpful assistant. Search information using the input that I give.

○ Knowledge 추가

▪ Add 버튼 클릭

Knowledge (0) + Add

 Knowledge gives the agent access to data sources for grounding responses. [Learn more](#)

▪ Grounding with Bing Search 클릭

Add knowledge ×

Expand the knowledge an agent has access to by connecting data sources it can use for grounding responses. Currently only a single instance per each type of data source is supported. [Learn more](#)

Add a data source

 Files Upload local files	 Azure AI Search Search and indexing	 Microsoft Fabric PREVIEW Accelerate data analysis with AI capabilities
 SharePoint PREVIEW Securely integrate and manage internal data	 Grounding with Bing Search Enhance model output with web data	 Grounding with Bing Custom... PREVIEW Enhance model output with data from selected web domains
 Tripadvisor Get travel data, guidance, and reviews	 Morningstar Access up-to-date analyst research, expert commentary, and essential Morningstar data	

- 이전에 생성한 Grounding with Bing Search connection 클릭 > Next

Choose an existing Grounding with Bing Search connection ×

← Back to select knowledge type

ⓘ When you use Grounding with Bing Search, your customer data is transferred outside of the Azure compliance boundary to the Grounding with Bing Search service. Grounding with Bing Search is not subject to the same data processing terms (including location of processing) and does not have the same compliance standards and certifications as the Azure AI Agent Service, as described in the [Grounding with Bing Search TOU](#). It is your responsibility to assess whether use of Grounding with Bing Search in your agent meets your needs and requirements.

[+ New connection](#) [Refresh](#)

Filter

Name	Authentication type	Added by	Key
✓ SOEUNPARKBINGSEARCH	API key	soeunpark@microsoft.c...	 🔍 🔑

[Configure Grounding with Bing Search optional parameters]

Count: 응답으로 리턴할 Search 결과 개수

Set language: 설정 언어(한국어의 경우 ko 입력)

Market: 검색 대상 국가 설정(한국의 경우 ko-KR 입력)

Freshness: 검색 결과 필터링에 사용할 날짜 지정

(2020-09-01~2025-08-31 입력 시 해당 날짜 사이의 결과만 반환)

- Try in playground 선택

Setup

🔍 Hide

🔍 Try in playground

Agent ID ⓘ

asst_Dje2ceAPsSowTvtVfmjPZpk7

Agent name

my-agent

Deployment *

[+ Create new deployment](#) ▼

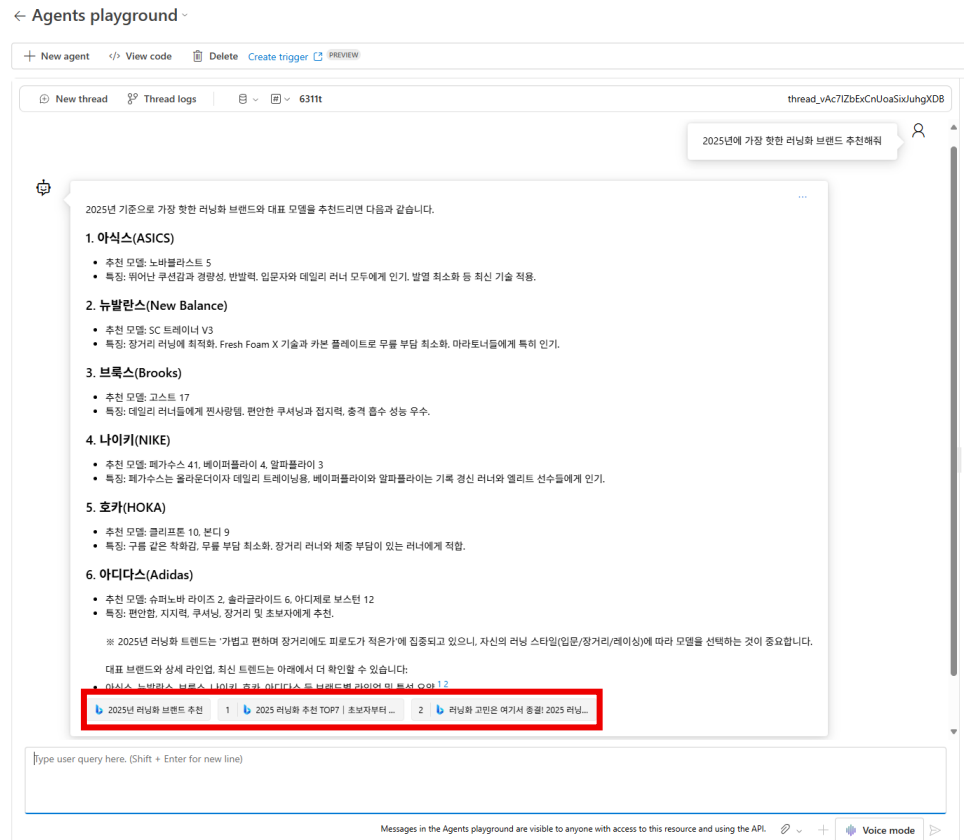
gpt-4.1 (version:2025-04-14)

Instructions ⓘ

You are a helpful assistant. Search for information using the input that I give.

- Try in playground 에서 생성한 agent 테스트

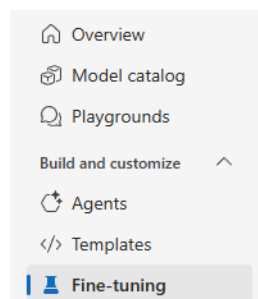
예) 2025 년에 가장 핫한 러닝화 브랜드 추천해줘



- Voice Agent (보여만 주기)
- Playground 내의 다른 기능 소개 (보여만 주기) – Speech playground > Fast transcription & Speech translation (whatstheweatherlike.wav 파일 사용)

7. 모델 Fine tuning

- 왼쪽 메뉴에서 Fine-tuning 선택



○ Fine tune model 버튼 클릭

Fine-tune with your own data

Adapt a pre-trained language model to excel at specific tasks or new domains by training it on targeted datasets, teaching the model new skills while retaining

[Generative AI fine-tuning](#)

AI Service fine-tuning

[+ Fine-tune model](#)

[Refresh](#)

[Deploy](#)

[Reset view](#)

○ gpt-4.1 선택 > Next

Select a model to fine-tune

Choose a model to fine-tune by filtering through our catalog. Note that specific models may only be available for fine-tuning in specific regions. [Learn more about regional constraints for fine-tuning](#)

Models 7Customization method

Show description

Search

Base modelsFine-tuned models

o4-mini

Chat completion

gpt-4.1

Chat completion

gpt-4.1-mini

Chat completion

gpt-4.1-nano

Chat completion

gpt-4o

Chat completion

gpt-4o-mini

Chat completion

gpt-4o-mini

Chat completion

gpt-4.1

Model version: 2025-04-14

Task: Chat completionSupervisedDirect Preference optimization

The gpt-4.1 series is the latest iteration of the `gpt-4o` model family. This iteration of models is specifically targeted for better coding and instruction following, making it better at handling complex technical and coding problems.

In addition, it increases the context token limit up to 1M input tokens and provides separate billing for small context (128k) and large (up to 1M) context inputs.

As with the previous gpt-4o model family, it supports a 16k output size and features such as:

- Text, image processing
- JSON Mode
- parallel function calling
- Enhanced accuracy and responsiveness
- Parity with English text and coding tasks compared to GPT-4 Turbo with Vision
- Superior performance in non-English languages and in vision tasks
- Support for enhancements
- Support for complex structured outputs.

Next

Cancel

○ Training data > Curated sample datasets > textGSM8K_train_sample

Fine-tune gpt-4.1

← Back to fine-tuning

Choose a dataset to customize your model. Ensure your training data is in a jsonl file, formatted for chat completions, with 'messages' as the row header.

Get started faster - learn how to create datasets using these samples ([chat data](#) , [multi-modal data](#)) on our [GitHub repo](#)

Training data *

Curated sample datasets

* Curated Sample Datasets GPT-4.1

Get started with a pre-prepared sample dataset to help you understand how to format data for fine-tuning and test end-to-end workloads. These datasets are intended as examples; they are not intended to produce production quality models.

Select data *

textGSM8K_train_sample

☒ Skip preview

[Learn more about preparing your own data](#)

Apply

Cancel

※Fine tuning 과정은 시간이 오래 소요될 수 있어서, 실습이 끝날 때까지 학습이 완료되지 않는다면
강제 종료를 부탁드립니다.

- Completed 확인

Fine-tune with your own data

Adapt a pre-trained language model to excel at specific tasks or new domains by training it on targeted datasets, teaching the model new skills while retaining it's general language

Generative AI fine-tuning AI Service fine-tuning

+ Fine-tune model Refresh Deploy Continual fine-tuning Delete Pause Cancel Reset view				
Search				
Model name	Base model	Status	Customization method	Created on
gpt-4.1-2025-04-14.ft-6cbf87f33cf347b890bb866decba61c7	gpt-4.1-2025-04-14	Completed	Supervised	Aug 31, 2025

- 배포된 모델 클릭하여 Details 확인

← [gpt-4.1-2025-04-14.ft-6cbf87f33cf347b890bb866decba61c7](#)

Details Metrics Logs Checkpoints

▶ Use this model Refresh Continual fine-tuning Delete Pause Cancel

Model attributes

ID
ftjob-6cbf87f33cf347b890bb866decba61c7

Status
✔ Completed

Created on
Aug 31, 2025 9:00 PM

Validation file
-

Method of Customization
Supervised

Auto-deployment enabled
true

Base model
gpt-4.1-2025-04-14

Training file
textGSM8K_train_sample.jsonl

Azure OpenAI resource
SOEUNPARK-FOUNDRY-EU2

Training type
Global (Preview)

Deployment type
Global standard

Training completed on
Aug 31, 2025 10:54 PM

Duration
1h 54m 33s

Training tokens billed
251,000

Deployment Status
[Deployment Successfully Triggered - View Details](#)

Download results Download training file Download validation file

Task parameters

Batch size
1

Learning rate multiplier
2

Number of epochs
3

Seed
367147176

- Fine tuning 된 모델 테스트

- [Models + endpoints] 탭 접속
- Open in playground 클릭

■ 테스트 질문 입력

예) 4 명의 학생 중 달리기 선수 2 명을 뽑는 경우의 수는 얼마야?

Chat history

4명의 학생 중 달리기 선수 2명을 뽑는 경우의 수는 얼마야?

Azure AI Foundry | gpt-4.1-2025-04-14.ft-6cbf87f33cf347b890bb866decba61c7

AI-generated content may be incorrect

4명 중 달리기 선수 2명을 뽑는 경우의 수는 조합(Combination)을 사용해 구합니다.

공식: $nCk = n! / (k! \times (n-k)!)$

여기서, $n=4$, $k=2$

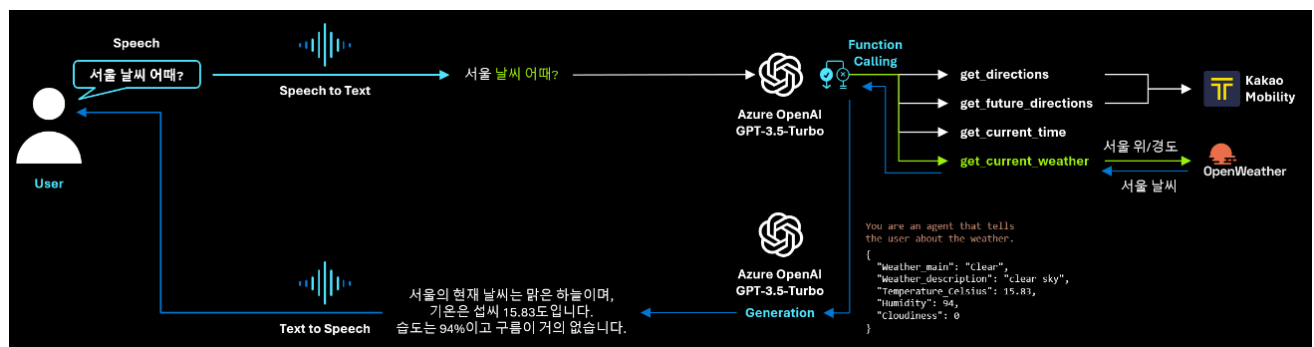
따라서 $4C2 = 4! / (2! \times 2!) = (4 \times 3 \times 2 \times 1) / ((2 \times 1) \times (2 \times 1)) = 24 / 4 = 6$

답: 6가지 경우가 있습니다.

#2 나만의 자비스 생성

OpenAI ChatGPT 의 Function Calling 기능을 활용하여, 아이언맨의 자비스와 같은 인공지능 비서 생성

- 사용자의 음성은 Azure 가 제공하는 AI Speech 서비스의 STT(Speech-to-text) 기능을 이용하여 텍스트로 변환
- 텍스트는 ChatGPT 가 제공하는 Function Calling 기능을 활용하여, 이미 작성되어진 API 또는 코드를 선택적으로 수행
- 수행 결과는 다시 Azure 가 제공하는 AI Speech 서비스의 TTS(Text-to-speech)를 이용하여 답변

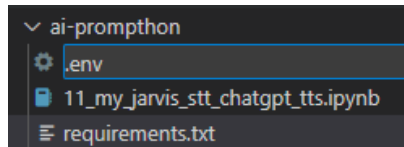


1. 환경 변수 설정 (.env)

- Explorer 탭에서 New File 아이콘 선택



- .env 생성

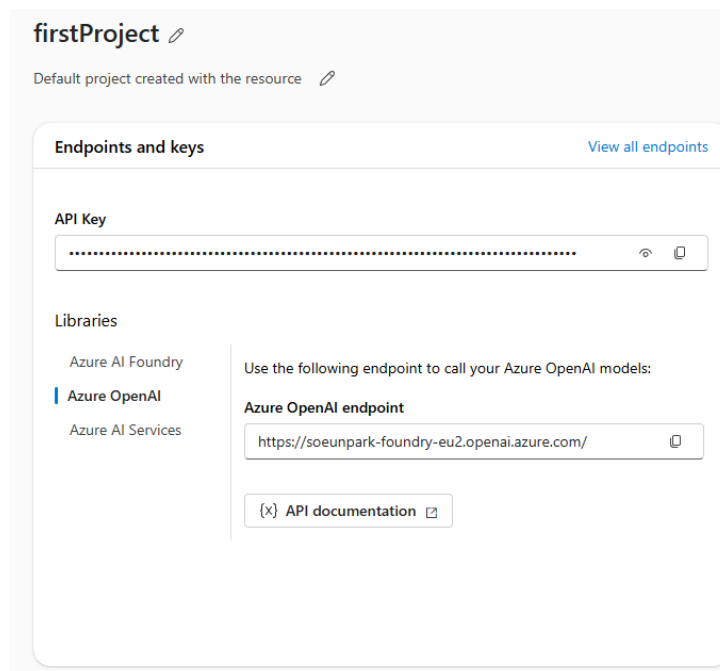


- 아래 내용 .env 에 복사 후 저장

```
AZURE_OPENAI_ENDPOINT=<OPENAI_ENDPOINT>
AZURE_OPENAI_API_KEY=<OPENAI_API_KEY>
OPENAI_API_VERSION=2025-03-01-preview
DEPLOYMENT_NAME=gpt-4.1
AZURE_SPEECH_KEY=<AI_SERVICES_API_KEY>
AZURE_SPEECH_REGION=eastus2
AZURE_SPEECH_LANGUAGE=ko=KR
KAKAO_REST_API_KEY=<KAKAO_REST_API_KEY>
WEATHER_API_KEY=<WEATHER_API_KEY>
```

※AZURE_OPENAI_API_KEY 와 AZURE_SPEECH_KEY 는 같은 값을 사용

- .env 값 가져오기 (Azure AI Foundry portal)



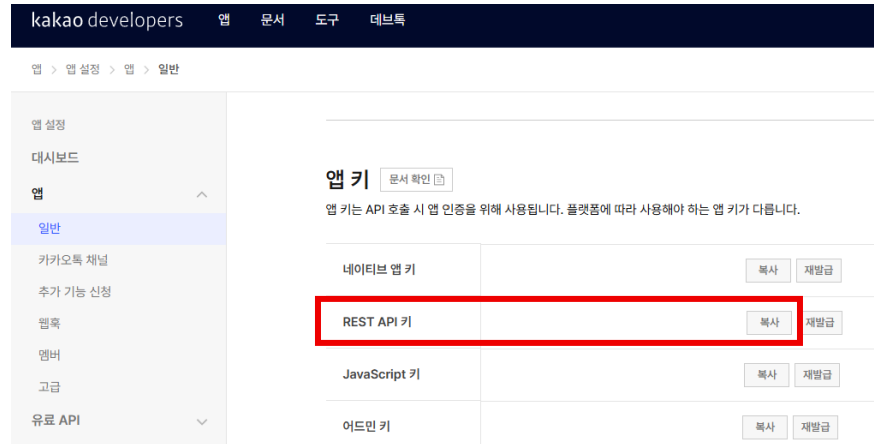
- KAKAO_REST_API_KEY 발급 방법

[REST API | Kakao Developers 문서](#)

- 앱 > 앱 생성



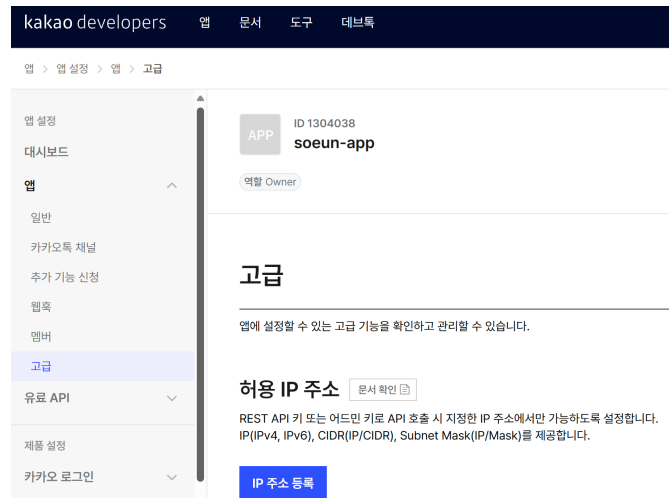
- 생성된 앱 클릭 > 앱 > 일반 > REST API 키 복사



- 허용 IP 주소 등록을 위해 네이버에서 내 IP 찾기



- 앱 > 고급 > 허용 IP 주소 등록 (REST API 키를 사용하려면 IP 주소 지정 필요)



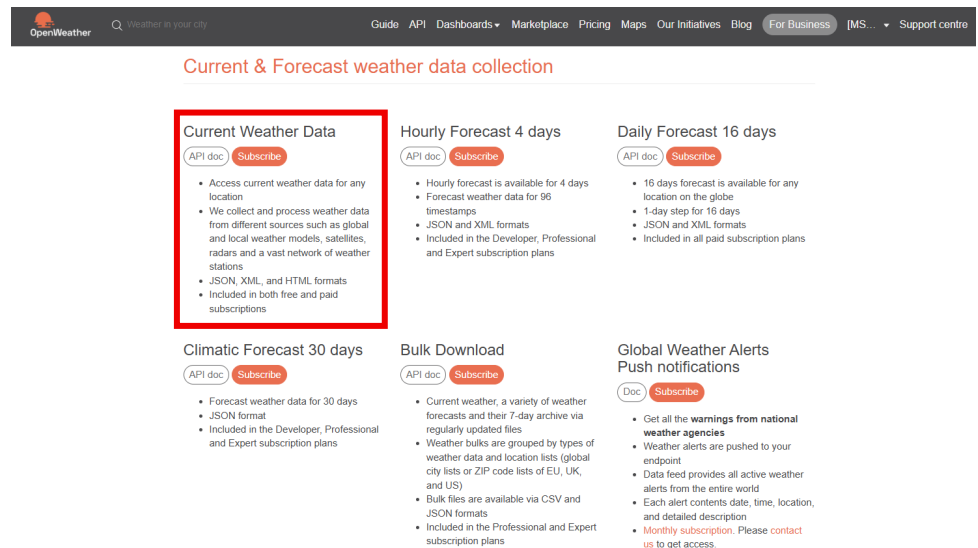
- 제품 설정 > 카카오맵 > 사용 설정 ON 으로 변경



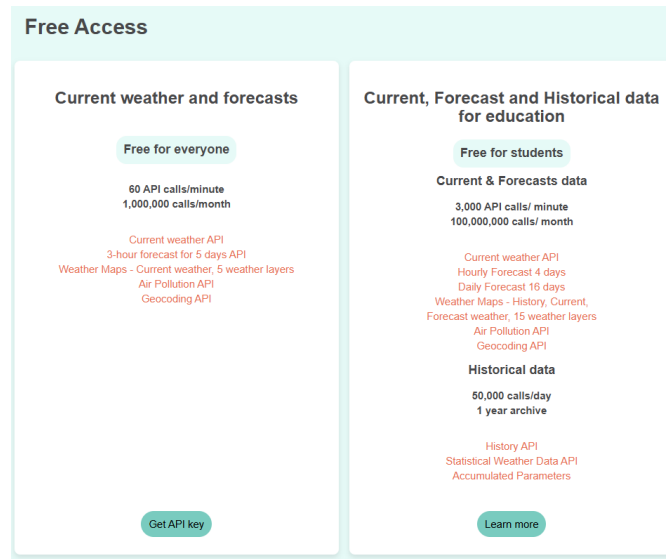
- WEATHER_API_KEY 발급 방법

Current weather data - OpenWeatherMap

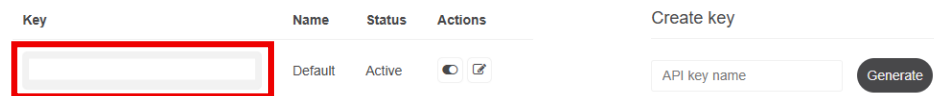
- 회원가입 > 메일인증필요
- API > Current Weather Data > Subscribe



- 스크롤을 아래로 쪽 내려서 Free Access > Get API key



- Key 값 복사

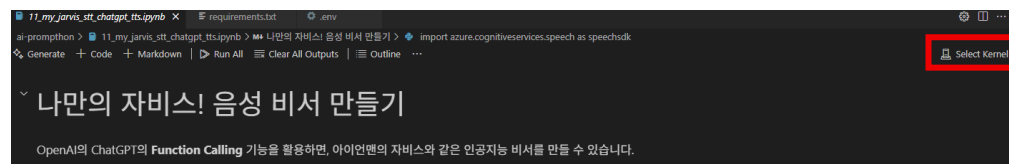


2. Jupyter notebook 코드 다운받기

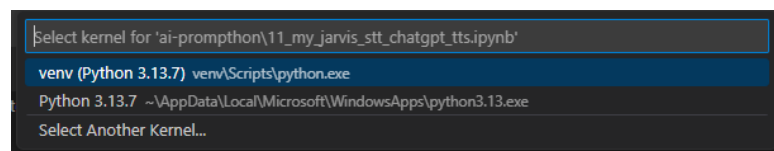
[azure-openai-samples-kr/quick start/11_my_jarvis_stt_chatgpt_tts.ipynb at main · HyunsooKim/azure-openai-samples-kr](https://github.com/HyunsooKim/azure-openai-samples-kr/blob/main/11_my_jarvis_stt_chatgpt_tts.ipynb)

3. VS Code 에서 코드 실행

- 다운로드 받은 Jupyter notebook 선택 > 오른쪽 위에 Select Kernal 클릭



- 이전에 생성한 Python venv 선택



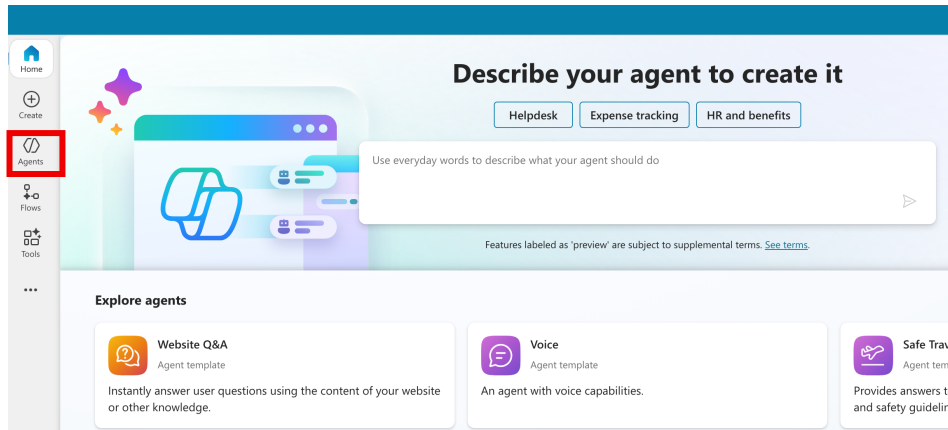
- Shift + Enter 눌러서 코드 하나씩 실행 후 결과 확인

※실행 초반에 ipykernel 설치 팝업이 뜨면 설치 진행

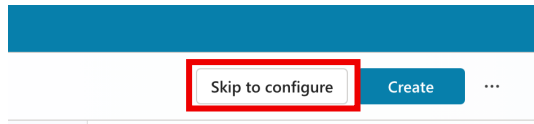
#3 AI Foundry with Copilot Studio 실습

AI Foundry 모델을 사용하는 Copilot Agent 만들기

1. Copilot Studio 접속 (copilotstudio.microsoft.com)
2. Agents > New agent

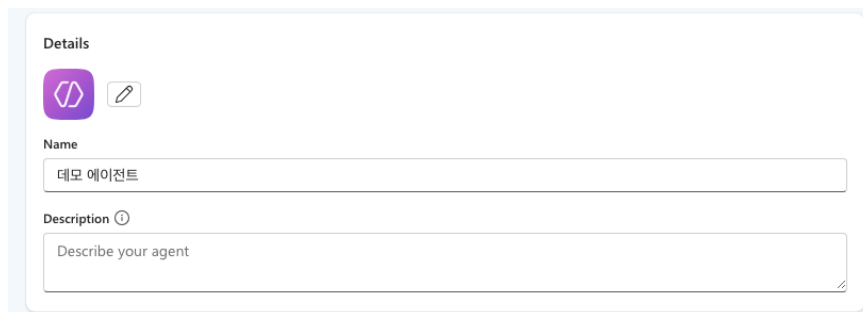


3. Copilot agent 가 실행되면 오른쪽 위에서 Skip to configure 선택



4. Agent 구성에서 다음과 같이 입력 후 Create

Name: 데모 에이전트



Details

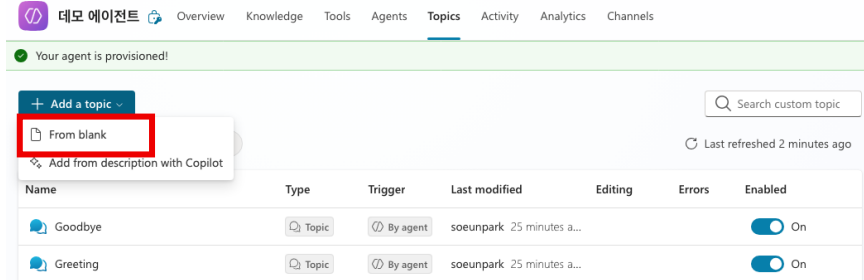
Name

데모 에이전트

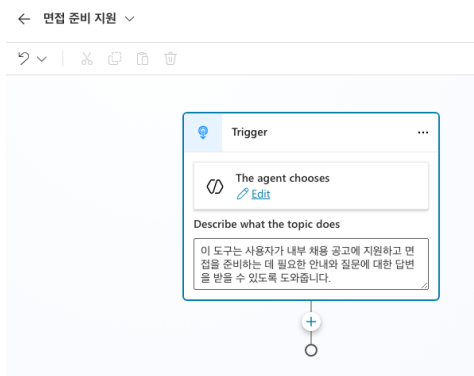
Description ⓘ

Describe your agent

5. 생성된 Agent 들어가서 Topics > Add a topic > From blank



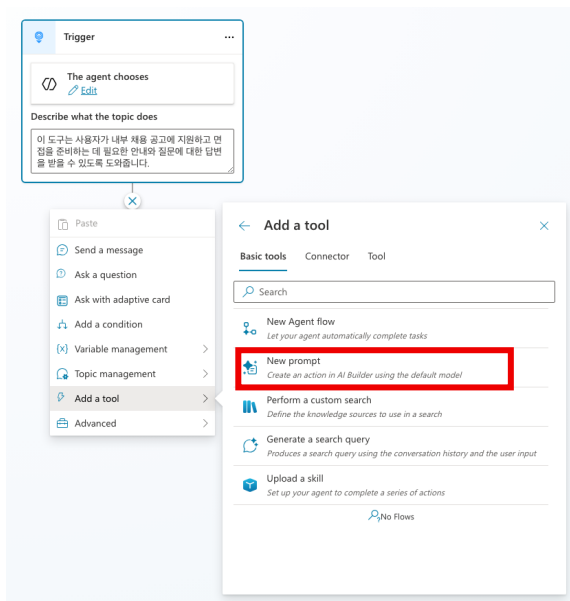
6. 토픽 정보 입력



Name: 면접 준비 지원

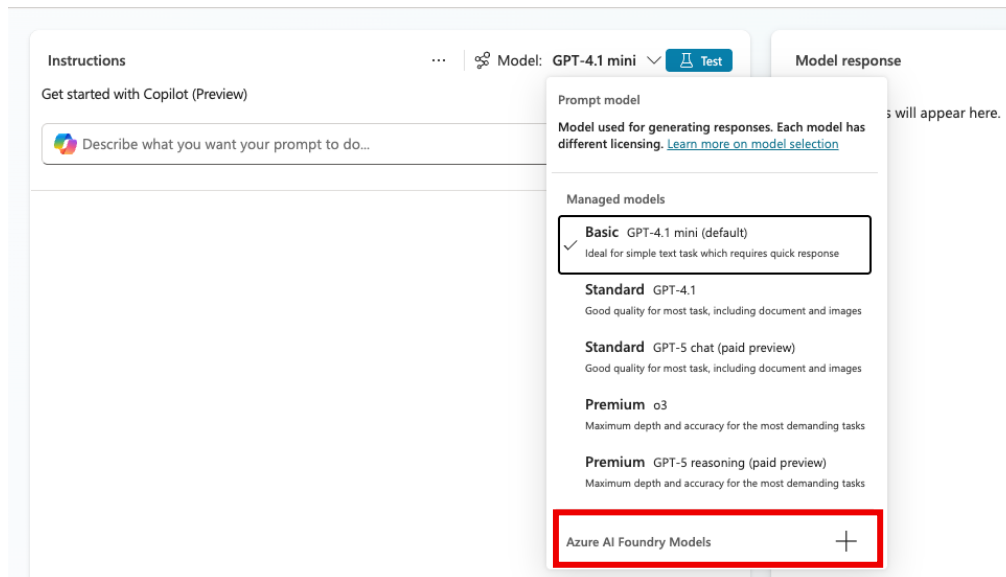
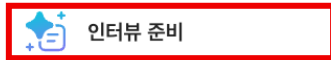
Describe what the topic does: 이 도구는 사용자가 내부 채용 공고에 지원하고 면접을 준비하는 데 필요한 안내와 질문에 대한 답변을 받을 수 있도록 도와줍니다.

7. 토픽이 생성되면 + > Add a tool > New prompt



8. 프롬프트 이름 변경

9. 모델 선택 > Azure AI Foundry Models



10. AI Foundry portal 에서 아래 필요한 정보 가져오기

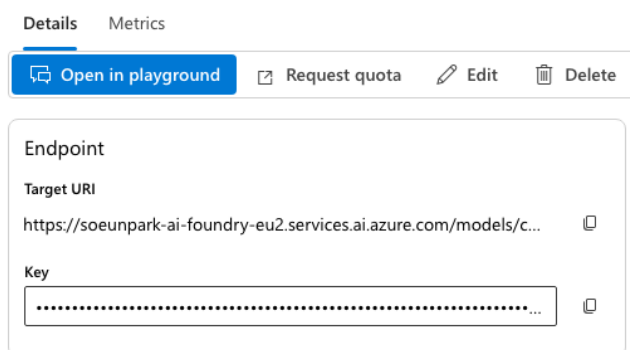
Model deployment name: <YOUR-MODEL-DEPLOYMENT-NAME>

Base model name: <YOUR-BASE-MODEL-NAME>

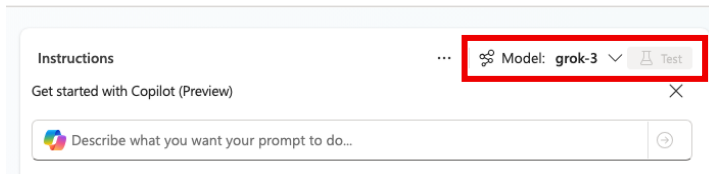
Azure model endpoint URL: <https://<PROJECT-NAME>cognitiveservices.azure.com/openai/deployments/<YOUR-MODEL-DEPLOYMENT-NAME>/chat/completions?api-version=2025-01-01-preview>

API Key: <YOUR-API-KEY>

← grok-3



11. 모델 설정 완료 확인



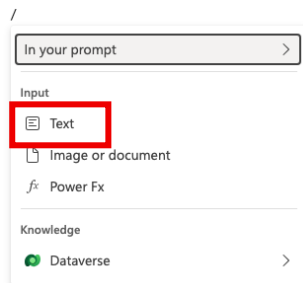
12. Instructions 을 아래와 같이 입력

내부 채용 면접 준비를 돕는 에이전트로서 /질의 에 응답하세요. 당신은 사용자가 내부 채용에 지원할 수 있도록 준비를 돕기 위해 여기에 있습니다.

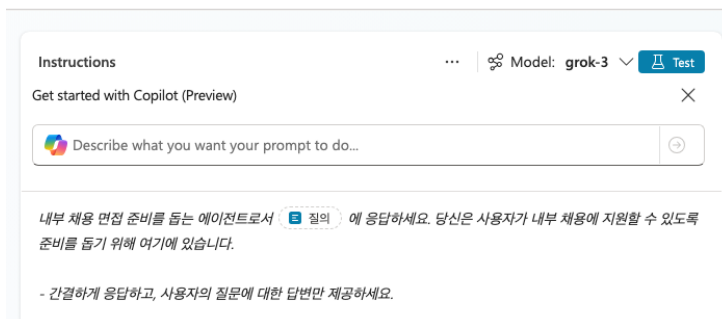
- 간결하게 응답하고, 사용자의 질문에 대한 답변만 제공하세요.

○ /질의 작성 방법

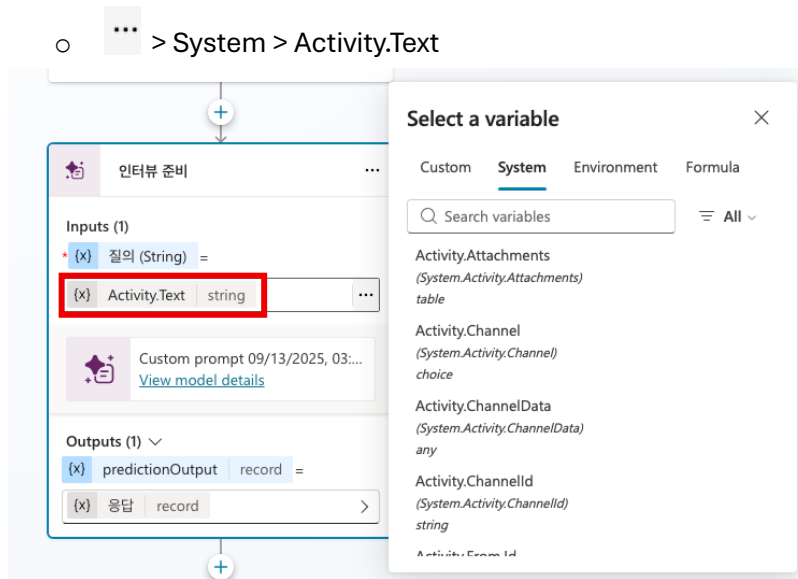
- / 입력 > Input 창 > Text > 질의 입력



13. 설정 확인

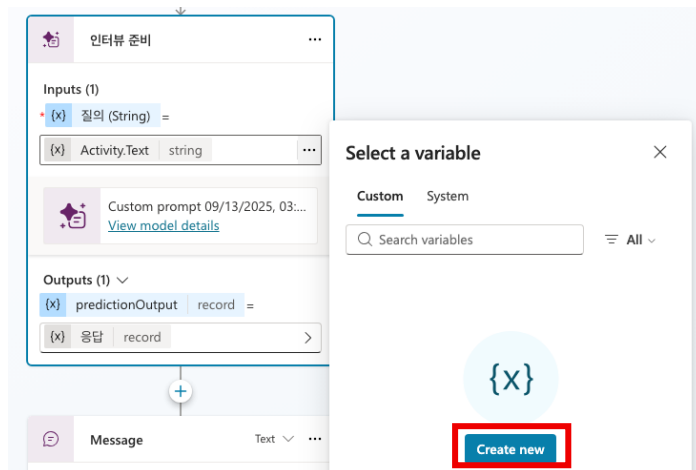


14. Inputs variable 추가



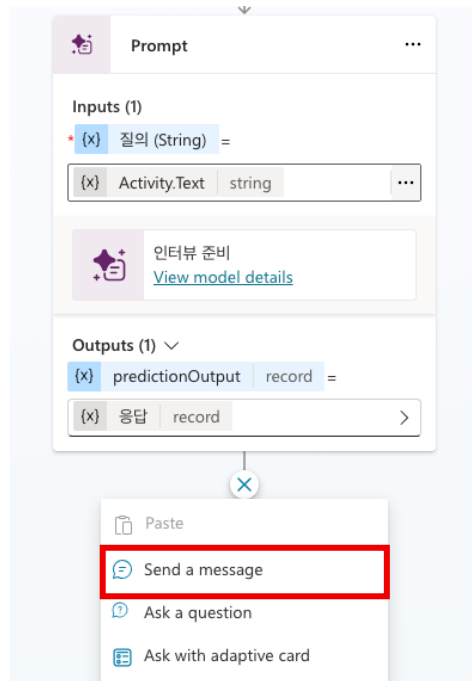
15. Outputs variable 추가

- Custom > Create new > 생성된 variable 클릭 > Variable name: 응답

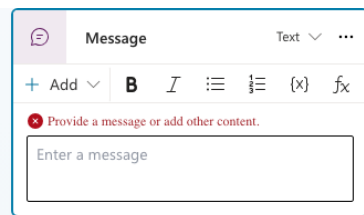


16. Message 추가

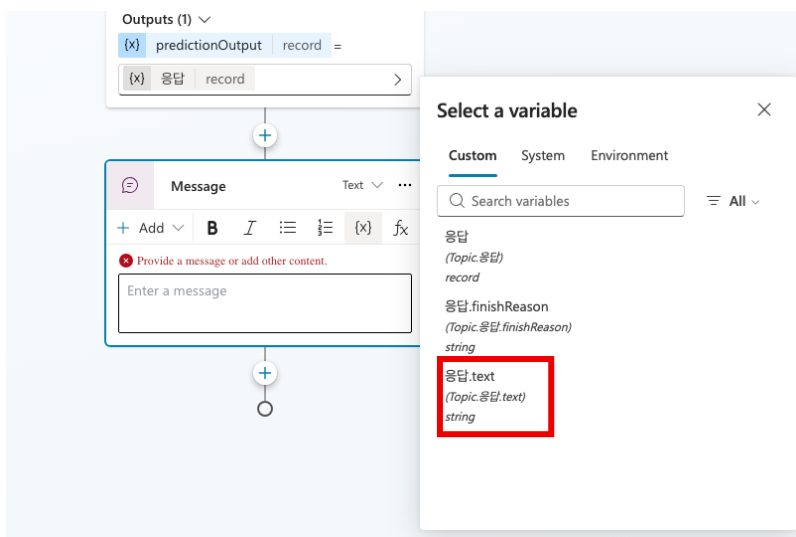
- + > Send a message



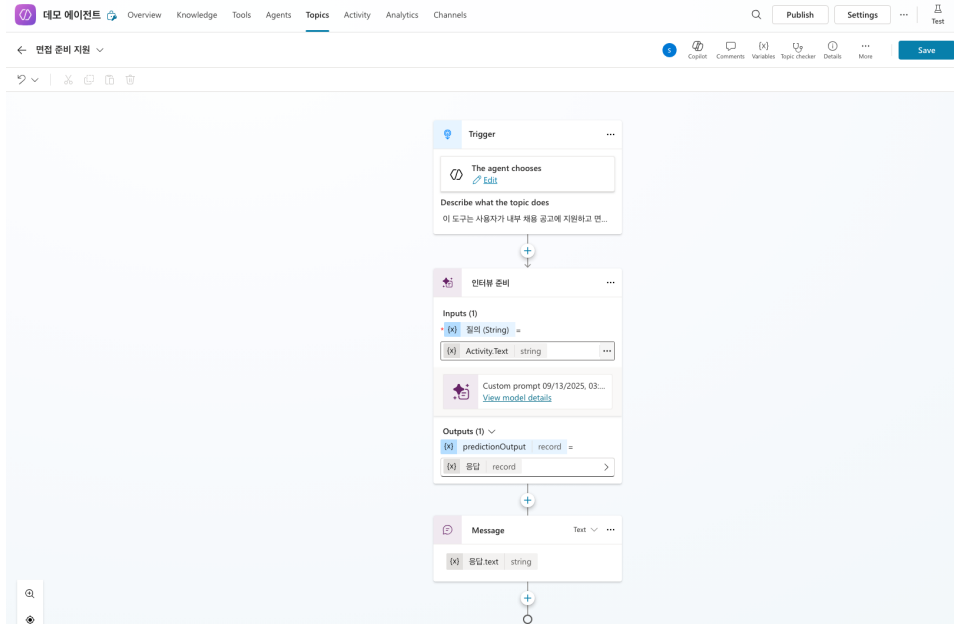
- {x} 선택



- Custom > 응답.text



17. Topic 저장



18. 에이전트 테스트

Test your agent

Hello, I'm 데모 에이전트, a virtual assistant. Just so you are aware, I sometimes use AI to answer your questions. If you provided a website during creation, try asking me about it! Next try giving me some more knowledge by setting up generative AI.

24 minutes ago

AI 기술 면접 질문 몇 개만 뽑아줘

Sure! Here are some AI technical interview questions that you might find useful:

- General AI Concepts:**
 - What is the difference between supervised, unsupervised, and reinforcement learning?
 - Explain the concept of overfitting and how to prevent it.
 - What is a confusion matrix, and how is it used?
 - Describe the bias-variance tradeoff.
- Machine Learning Algorithms:**
 - How does a decision tree algorithm work?
 - Explain the working of a support vector machine (SVM).
 - What is the k-nearest neighbors (KNN) algorithm, and how does it work?
 - Describe the working of a random forest algorithm.
- Neural Networks and Deep Learning:**
 - What is a neural network, and how does it learn?
 - Explain the concept of backpropagation.
 - What are convolutional neural networks (CNNs), and where are they used?
 - Describe recurrent neural networks (RNNs) and their applications.

19. 배포

